



Solving complex linear equations

Task A: Introduction to complex linear equations

1) Alex solved this complex linear equation.
Fill in the missing blanks in his work.



$$\frac{7x - 5}{8} = 2$$

$$\frac{7x - \boxed{}}{8} = \frac{\boxed{}}{8}$$

$$7x - \boxed{} = \boxed{}$$

$$7x = \boxed{}$$

$$x = \boxed{}$$

$$\frac{2}{1} = \frac{\boxed{}}{8}$$

2) Solve:

a) $\frac{2x + 10}{3} = 14$

b) $\frac{2x}{3} + 10 = 14$

Task B: Solving complex linear equations

1) Where is the error in this attempted solution?

Find the error and write a sentence explaining why it is wrong.

$$\frac{3x - 5}{2} = \frac{2x + 1}{10}$$

$$\frac{3x - 5}{2} \times \frac{1}{5} = \frac{2x + 1}{10}$$

$$\frac{3x - 5}{10} = \frac{2x + 1}{10}$$

$$3x - 5 = 2x + 1$$

$$x - 5 = 1$$

$$x = 6$$

2) Aisha successfully solved two equations. Fill in the missing blanks in her working.



$$\begin{aligned}\frac{3x-5}{2} &= \frac{2x+1}{10} \\ \frac{3x-5}{2} \times \boxed{} &= \frac{2x+1}{10} \\ \boxed{}(3x-5) &= \frac{2x+1}{10} \\ \boxed{}(3x-5) &= 2x+1 \\ \boxed{} - \boxed{} &= 2x+1 \\ \boxed{} &= 26 \\ x &= 2\end{aligned}$$

$$\begin{aligned}\frac{5-x}{3} &= \frac{3-x}{5} \\ \frac{5-x}{3} \times \frac{5}{5} &= \frac{3-x}{5} \times \boxed{} \\ \frac{5(5-x)}{15} &= \frac{\boxed{}(3-x)}{15} \\ 5(5-x) &= \boxed{}(3-x) \\ 25-5x &= \boxed{} - \boxed{} \\ 16 &= \boxed{} \\ x &= \boxed{}\end{aligned}$$

Task C: Solving simple rational equations

1) Finish each method to solve this equation in two different ways:

$$\begin{aligned}\frac{54}{e} &= 6 \\ e\left(\frac{54}{e}\right) &= e(6)\end{aligned}$$

$$\begin{aligned}\frac{54}{e} &= 6 \\ \frac{54}{e} &= 6 \times \frac{e}{e}\end{aligned}$$

2) Solve:

a) $\frac{84}{x} = 12$

b) $11 = \frac{84}{x}$

c) $-7 = \frac{84}{3x}$

